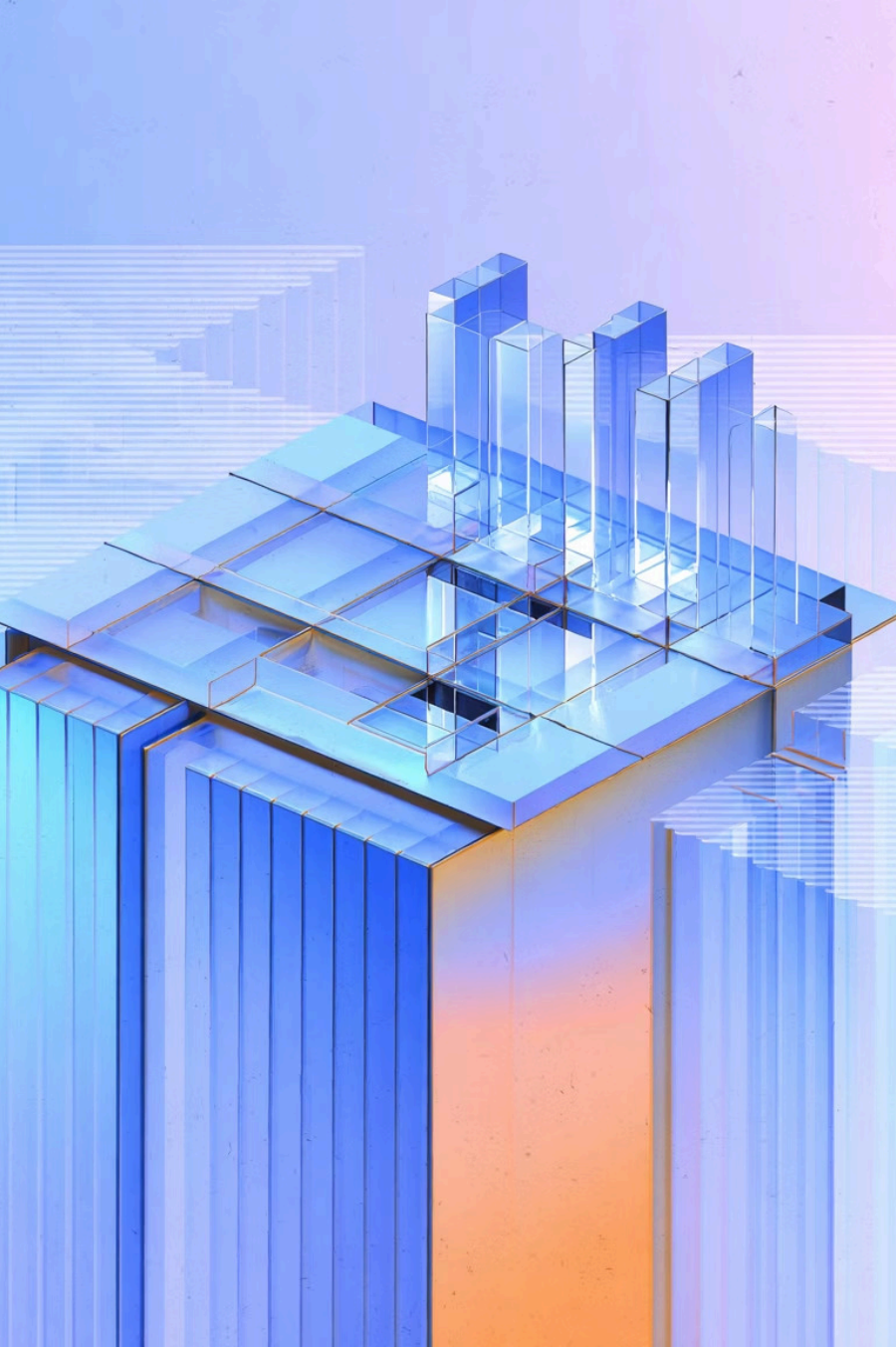




Customer Shopping Behavior Analysis

3,900 purchases • 18 columns • goal: uncover spending patterns, segments, preferences, subscriptions



Dataset Summary

Rows & Columns

3,900 rows · 18 columns

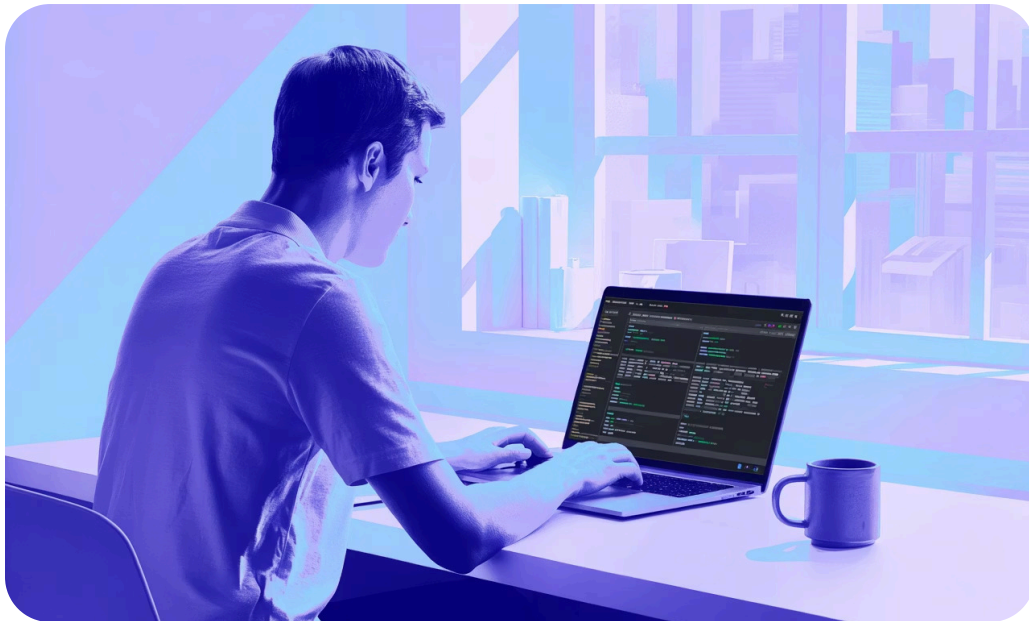
Key Features

Demographics, purchase details, behavior

Missing Data

37 nulls in Review Rating

Data Preparation (Python)



01

Load & Explore

pandas: `df.info()`, `df.describe()`

02

Clean & Impute

Median imputation for Review Rating

03

Feature Eng.

`age_group`, `purchase_frequency_days`

04

DB Integration

Loaded cleaned DataFrame to PostgreSQL

Dataset Stats (Highlights)



Average Purchase

\$59.76



Average Rating

3.75 (mean)



**Avg Previous
Purchases**

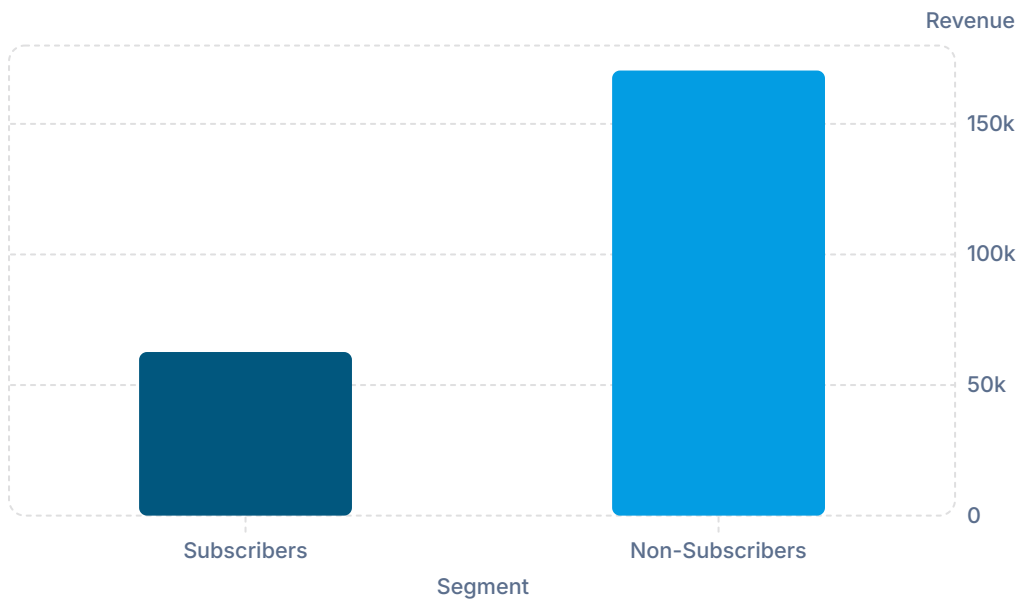
25.35



Age (mean)

44.07 yrs

Revenue & Subscriptions (SQL)

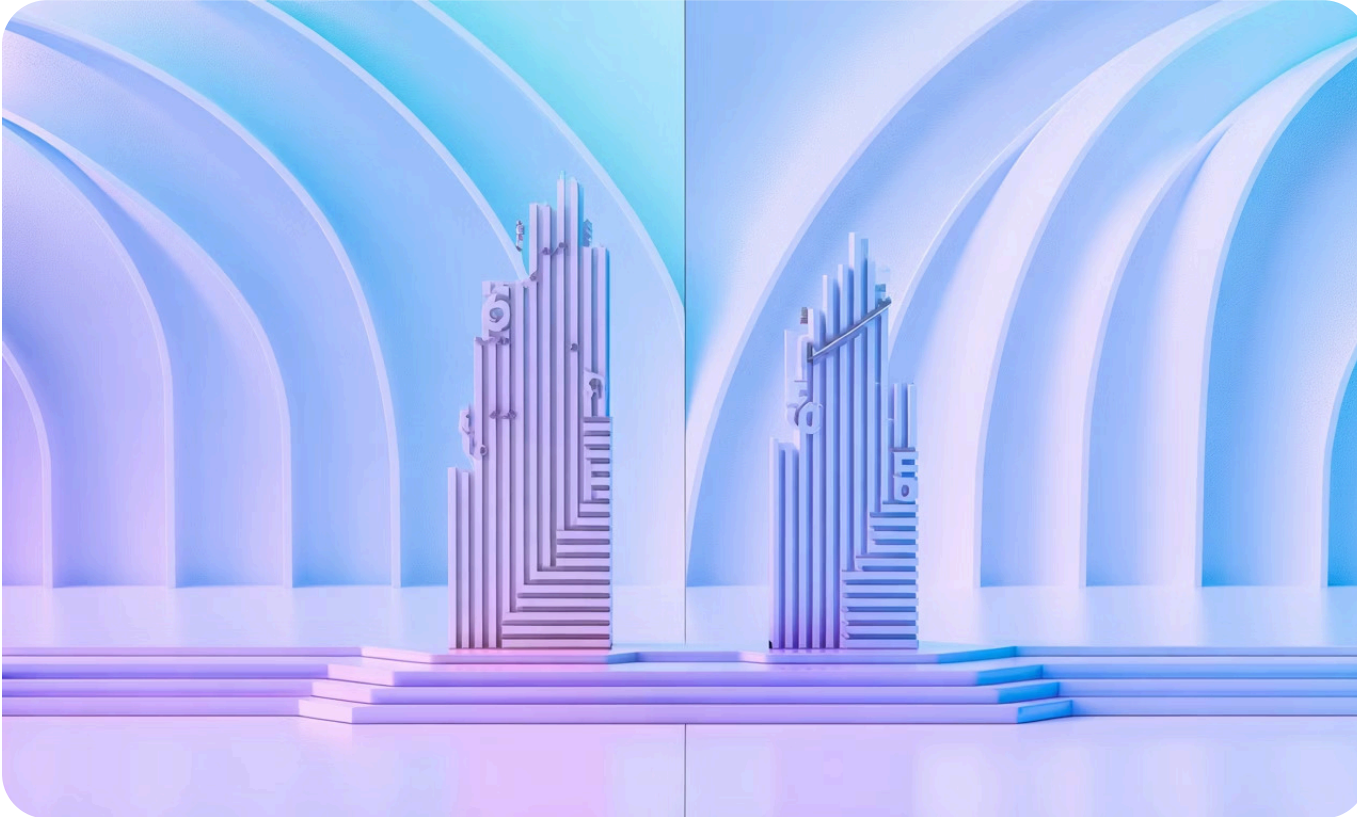


Subscribers: 1,053 customers · avg spend \$59.49 · total \$62,645

Non-Subscribers: 2,847 customers · avg spend \$59.87 · total \$170,436

i Recommend boosting subscriptions — close gap in total revenue

Revenue by Gender



Female revenue: 75,191

Male revenue: 157,890

Insight: male segment drives higher revenue – tailor campaigns accordingly

Top Products & Categories



Gloves

Top rated



Sandals

Top rated



Boots

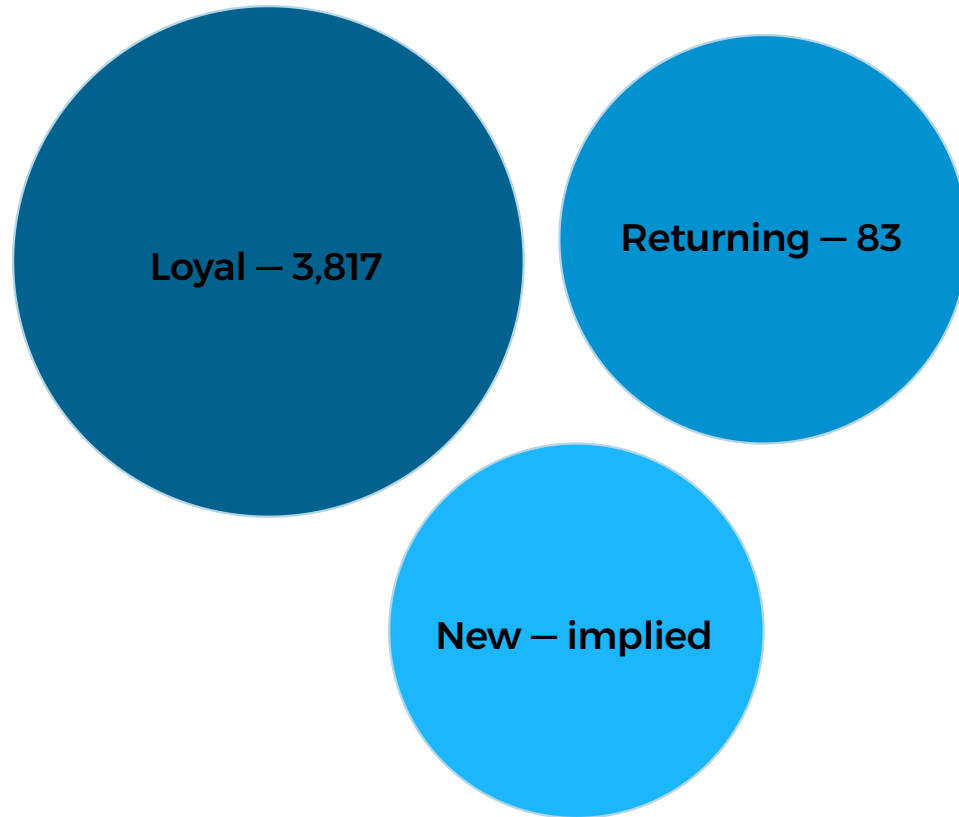
Top rated



Clothing

Best-selling items: Blouse,
Pants, Shirt

Customer Segments & Loyalty

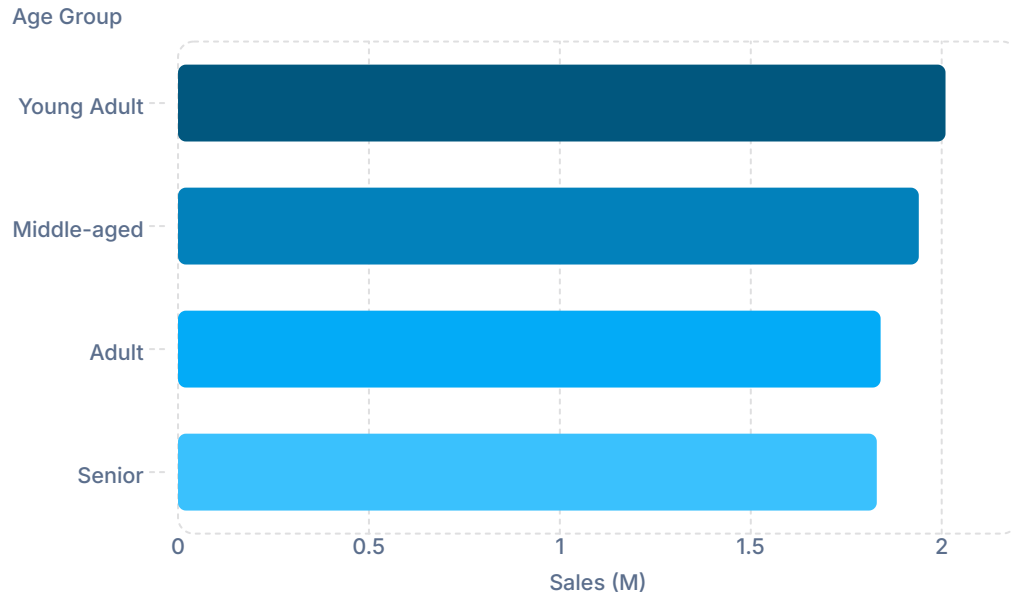


Repeat buyers (>5 purchases): more likely to subscribe – Yes: 958 · No: 2,518

- ✓ Build loyalty program to convert repeat buyers into Loyal segment

Majority classified as Loyal (3,817); Returning: 83

Age Groups: Sales & Revenue



Revenue by Age Group:

- Young Adult: 62,143
- Middle-aged: 59,197
- Adult: 55,978
- Senior: 55,763

📌 Target high-revenue age groups with tailored offers

Recommendations & Dashboard

Power BI snapshot:



KPIs: 3.9K customers · \$59.76 avg purchase · 3.75 avg rating